Piper PA-44-180 Seminole (PA44)

2× Lycoming O-360-A1H6 / LO-360-A1H6 (counter-rotating), 180 hp each · 4 seats · Typical Seminole (1979+ and 1995+ production) POH procedures; carbureted counter-rotating engines, manual flaps, 3,800 lb MTOW. Unfeathering accumulators where fitted.

V-speeds & limits

Vr	**75 KIAS**
Vmc	**56 KIAS**
Vsse	**82 KIAS**
Vxse	**82 KIAS**
Vyse	**88 KIAS**
Vx	**82 KIAS**
Vy	**88 KIAS**
Best glide	**105 KIAS (both engines out)**
Vfe	**111 KIAS**
Vle	**140 KIAS**
Vlo	**109 KIAS retract / 140 KIAS extend**
Va	**135 KIAS (3,800 lb) / 112 KIAS (2,700 lb)**
Vno	**169 KIAS**
Vne	**202 KIAS**
Vs / Vso	**57 / 55 KIAS**
Approach	**75 KIAS final (flaps 40°); 90 KIAS single-engine**

Normal procedures

Before Start

- Preflight inspection ..... **COMPLETE**
- Fuel quantity ..... **CHECK visually (54 gal usable per side, 108 total)**
- Baggage / nose doors ..... **CLOSED & LOCKED**
- Passenger brief ..... **COMPLETE (belts, door, exits, sterile cockpit)**
- Seats, belts, harnesses ..... **ADJUST & LOCK**
- Door ..... **CLOSED & LATCHED (top and side)**
- Parking brake ..... **SET**
- Gear selector ..... **DOWN**
- Avionics master ..... **OFF**
- Electrical switches ..... **OFF**
- Circuit breakers ..... **CHECK IN**
- Alternate static source ..... **OFF (closed)**
- Fuel selectors ..... **ON (both)**
- Cowl flaps ..... **OPEN**
- Alternators ..... **ON (both)**
- Carb heat ..... **OFF (both)**
- Propeller controls ..... **FULL FORWARD**

Engine Start

Master switch	**ON**
Gear lights	**THREE GREEN**
Beacon	**ON**
Mixtures	**FULL RICH**
Throttles	**OPEN 1/2 inch**
Electric fuel pumps	**ON**
Primer	**PRIME 0–3 strokes (as required), LOCKED**
Propeller area	**CLEAR**
Starter (left)	**ENGAGE (10 s max, then 30 s cool-down)**
Throttle	**ADJUST 800–1,000 RPM**
Oil pressure	**CHECK rising within 30 s (60–90 psi green)**
Starter (right)	**ENGAGE, repeat for right engine**
Oil pressure (right)	**CHECK rising within 30 s**
Alternators / ammeters	**CHECK charging (both)**
Electric fuel pumps	**OFF (fuel pressure check 0.5–8 psi each)**
Avionics master	**ON**
Flaps	**RETRACT (check handle latches)**

Before Taxi

Flight instruments / avionics	**CHECK & SET**
Altimeters	**SET**
Heading indicator	**SET to compass**
Transponder	**STBY / as required**
Parking brake	**RELEASE**
Brakes	**CHECK during initial roll**
Steering	**CHECK (nosewheel steering with rudder pedals)**
Gyros	**CHECK during turns (turn coordinator, DG, attitude)**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Fuel selectors	**ON (both)**
Mixtures	**FULL RICH**
Propeller controls	**FULL FORWARD**
Throttles	**2,000 RPM (each engine in turn)**
Magnetos	**CHECK each engine (max drop 175 RPM, max difference 50 RPM)**
Propellers	**CYCLE 2–3 times (RPM drop approx 300–500, oil pressure steady), then FULL FORWARD**
Carb heat	**CHECK each (RPM drop), then OFF**
Alternators	**CHECK each (load on ammeter, turn one off at a time)**
Vacuum	**4.8–5.2 in Hg (both pumps, check each by annunciator)**
Throttles	**1,500 RPM**
Feather check	**PROP CONTROLS to FEATHER briefly, note RPM drop (approx 500), FORWARD before 1,000 RPM**
Oil pressure / temperature	**CHECK GREEN (60–90 psi)**
Fuel pressure	**CHECK GREEN (0.5–8 psi)**
Annunciator panel	**PRESS TO TEST**
Throttles	**IDLE check (500–700 RPM), then 1,000 RPM**
Electric fuel pumps	**ON**
Cowl flaps	**OPEN**
Flaps	**SET (0° normal / 25° short field)**
Trim	**SET for takeoff (neutral, rudder trim neutral)**
Gear lights	**THREE GREEN, gear selector DOWN**
Seat belts / harnesses	**FASTENED**
Door	**LATCHED (top and side)**
Pitot heat / lights	**AS REQUIRED**
Transponder	**ALT**
Takeoff briefing	**COMPLETE (abort below 75 KIAS; Vyse 88; gear-up point; engine failure plan)**

After Takeoff / Climb

Airspeed	**ROTATE 75 KIAS, accelerate to 88 KIAS (Vy / Vyse)**
Landing gear	**UP (positive rate, below 109 KIAS, no usable runway)**
Gear lights	**OUT, gear-unsafe light OUT**
Flaps	**RETRACT (if used, when clear of obstacles)**
Power	**FULL / 2,700 RPM to safe altitude, then 25 in Hg / 2,500 RPM**
Electric fuel pumps	**OFF (at safe altitude, check fuel pressure)**
Cowl flaps	**AS REQUIRED (open in climb)**
Airspeed	**105 KIAS cruise climb**
Mixtures	**FULL RICH (lean above 5,000 ft)**
Engine instruments	**CHECK**
Landing light	**OFF (as desired)**

Cruise

Power	**SET (2,300–2,500 RPM, manifold pressure per chart, 75% max)**
Props	**SYNCHRONIZE**
Mixtures	**LEAN (per power chart)**
Cowl flaps	**CLOSED (as required for CHT / oil temp)**
Fuel quantity / balance	**CHECK (crossfeed to balance if required, level flight only)**
Engine instruments	**CHECK**

Descent

Altimeters	**SET**
Mixtures	**ENRICHEN as altitude decreases**
Carb heat	**AS REQUIRED**
Power	**REDUCE gradually (keep manifold pressure above approx 15 in Hg)**
Cowl flaps	**CLOSED**
Fuel selectors	**ON (both, crossfeed OFF)**
Airspeed	**BELOW 140 KIAS before gear extension**
Approach briefing	**COMPLETE**

Before Landing

Seat belts / harnesses	**FASTENED**
Fuel selectors	**ON (both)**
Electric fuel pumps	**ON**
Mixtures	**FULL RICH**
Carb heat	**AS REQUIRED**
Propeller controls	**FULL FORWARD (on final)**
Landing gear	**DOWN (below 140 KIAS)**
Gear lights	**THREE GREEN, nose gear in mirror**
Cowl flaps	**AS REQUIRED**
Flaps	**SET (below 111 KIAS; 40° normal landing)**
Airspeed	**75 KIAS final (flaps 40°)**
Landing light	**ON**
Gear	**RECHECK THREE GREEN on final**

After Landing

Flaps	**RETRACT**
Electric fuel pumps	**OFF**
Carb heat	**OFF**
Cowl flaps	**OPEN**
Transponder	**STBY / as required**
Lights	**AS REQUIRED**

Shutdown

Parking brake	**SET**
Avionics master	**OFF**
Electrical switches	**OFF**
Propeller controls	**FULL FORWARD**
Throttles	**1,000 RPM**
Mixtures	**IDLE CUT-OFF**
Magnetos	**OFF (after props stop)**
Alternators	**OFF**
Master switch	**OFF**
Fuel selectors	**ON (leave on, per POH) / OFF per operator**
Control lock / chocks	**INSTALL**

Abnormal procedures

Gear Fails to Retract / Gear Unsafe Light

Red gear-unsafe light or hydraulic pump running continuously after selecting UP

Airspeed	**BELOW 109 KIAS (Vlo retract)**
Gear selector	**RECYCLE DOWN, three green, then UP**
Gear circuit breaker / pump breaker	**CHECK IN**

▶ Gear retracts, lights out, pump stops?

Yes:

Flight	**CONTINUE, monitor on next extension**

No:

Gear selector	**DOWN**
Gear lights	**THREE GREEN**
Airspeed	**BELOW 140 KIAS**
Hydraulic pump	**PULL breaker if running continuously**
Deferred: Gear left down – limit speed to 140 KIAS, reduced climb and range (single-engine climb poor). Land as soon as practical.	**NOTED**

Gear Warning Horn

Horn with gear up: throttle below approx 14 in Hg, or flaps beyond 10°

Gear position	**CHECK**

▶ Intentional gear-up (e.g. descent)?

Yes:

Horn	**EXPECT; extend gear before landing**
Gear	**DOWN before final, three green**

No – landing:

Landing gear	**DOWN, three green**

Alternator Failure (one or both)

ALT annunciator / ammeter zero

Ammeters	**CHECK which alternator has no output**
ALT circuit breaker (affected)	**CHECK, RESET once**
ALT switch (affected)	**OFF, then ON**

▶ Alternator restored?

Yes:

Ammeters	**MONITOR, balance load**

No – one alternator:

ALT switch (affected)	**OFF**
Electrical load	**REDUCE to within remaining alternator (approx 60 A)**
Deferred: One alternator failed – reduced electrical capacity; land as soon as practical.	**NOTED**

No – both alternators:

Both ALT switches	**OFF**
Electrical load	**MINIMUM (one radio, transponder)**
Landing gear	**EXTEND early; use emergency knob if battery is depleted**
Deferred: Both alternators failed – battery only (approx 30 min); extend gear early (free-fall if needed), expect no gear lights late in flight.	**NOTED**

Low Oil Pressure (one engine)

Indication below 25 psi at idle / below 60 psi in flight

Oil pressure / temperature (affected) ..... **CROSS-CHECK**

▶ Oil temperature rising with low pressure?

Yes – real:

- Power (affected) ..... **REDUCE**
- Engine (affected) ..... **SHUT DOWN and FEATHER before seizure (see Engine Securing)**
- Deferred: Engine shut down for low oil pressure – single-engine approach and landing. ..... **NOTED**

No – gauge suspect:

- Flight ..... **CONTINUE to nearest suitable airport, monitor**
- Deferred: Oil pressure indication unreliable on one engine – land as soon as practical. ..... **NOTED**

High Oil Temperature / CHT (one engine)

Indication above 245°F / in the red

Cowl flap (affected) ..... **OPEN**

Mixture (affected) ..... **ENRICHEN**

Power (affected) ..... **REDUCE**

Airspeed ..... **INCREASE (lower nose)**

Oil pressure ..... **CHECK**

▶ Temperature decreasing?

Yes:

- Flight ..... **CONTINUE, monitor**

No:

- Engine (affected) ..... **CONSIDER precautionary shutdown and feather**
- Deferred: Engine overheat – reduced power on affected engine; land as soon as possible. ..... **NOTED**

Rough Running / Carburetor Ice

RPM loss, rough engine

Carb heat (affected) ..... **ON (expect initial further roughness)**

Mixture (affected) ..... **ADJUST for smoothest running**

Electric fuel pump (affected) ..... **ON**

Fuel selector (affected) ..... **CHECK ON (try X-FEED)**

Magnetos (affected) ..... **CHECK L / R / BOTH (operate on best if one is faulty)**

▶ Engine smooth again?

Yes:

- Carb heat ..... **LEAVE ON if icing conditions**
- Deferred: Carb icing conditions – carb heat on, reduced performance. ..... **NOTED**

No:

- Power (affected) ..... **REDUCE to smoothest setting or shut down and feather**
- Deferred: Rough-running engine – be prepared to feather; land as soon as possible. ..... **NOTED**

Low Fuel Pressure (one engine)

Fuel pressure below 0.5 psi or fluctuating

Electric fuel pump (affected) ..... **ON**

Fuel quantity (affected side) ..... **CHECK**

Fuel selector (affected) ..... **X-FEED if that side is low**

Mixture (affected) ..... **FULL RICH**

▶ Pressure restored?

Yes:

- Electric fuel pump ..... **LEAVE ON**
- Deferred: Electric fuel pump on continuously on one engine – engine-driven pump suspect. Land as soon as practical. ..... **NOTED**

No:

- Engine (affected) ..... **EXPECT failure – prepare to feather**

See Engine Failure in Flight.

Propeller Overspeed

RPM above 2,700 on one engine and not responding to prop control

Throttle (affected) ..... **RETARD**

Oil pressure (affected) ..... **CHECK**

Propeller control (affected) ..... **FULL DECREASE RPM, then set if any control remains**

Airspeed ..... **REDUCE**

Throttle (affected) ..... **AS REQUIRED to keep RPM at or below 2,700**

▶ RPM controllable?

Yes:

Deferred: Prop governor fault – control RPM with throttle; land as soon as practical. ..... **NOTED**

No:

Engine (affected) ..... **FEATHER and SECURE (if feather works)**

Deferred: Engine shut down for prop overspeed – single-engine approach and landing. ..... **NOTED**

Flap Handle Will Not Latch / Flaps Inoperative

Flap handle ..... **RETRY, hold then release latch slowly**

▶ Flaps hold a setting?

Yes:

Flaps ..... **USE AVAILABLE setting**

No:

Flaps ..... **UP (handle stowed)**

Approach ..... **85–90 KIAS, flaps-up landing**

Deferred: Flaps inoperative – flaps-up landing at 85–90 KIAS, longer landing distance. ..... **NOTED**

Door Open in Flight

Airspeed ..... **SLOW to approx 90 KIAS**

Storm window ..... **OPEN (cabin vents closed)**

Door ..... **PULL CLOSED, latch side then top**

▶ Door latched?

Yes:

Flight ..... **CONTINUE**

No:

Aircraft is controllable with door unlatched. Expect noise and some yaw.

Deferred: Door unlatched – land normally, secure door on the ground. ..... **NOTED**

Vacuum Pump Failure

Vacuum annunciator (L or R) / vacuum below 4.8 in Hg

Vacuum gauge / annunciators ..... **CHECK which pump failed**

▶ One pump or both?

One pump:

Gyros ..... **CHECK operating normally on remaining pump**

Deferred: One vacuum pump failed – gyros on single pump; avoid extended IMC. ..... **NOTED**

Both pumps:

Attitude indicator / DG ..... **CROSS-CHECK then COVER**

IMC ..... **EXIT, request no-gyro assistance if needed**

Deferred: Vacuum failed – no attitude indicator / DG. Avoid IMC; partial-panel landing. ..... **NOTED**

Static Source Blocked

Airspeed / altimeter / VSI erratic

Pitot heat ..... **ON**

Alternate static source ..... **OPEN (valve under left side of panel)**

Storm window / vents ..... **CLOSED (for calibration accuracy)**

Airspeed and altimeter read slightly high with alternate static; see POH correction table.

Deferred: Alternate static in use – small airspeed/altimeter errors; gear/flap speeds by indication. ..... **NOTED**

Inadvertent Icing

Pitot heat ..... **ON**

Carb heat ..... **ON (both)**

Cabin heat / defroster ..... **ON (windshield)**

Exit icing conditions ..... **TURN / CLIMB / DESCEND**

Airspeed ..... **INCREASE, avoid abrupt maneuvers**

Flaps ..... **LIMIT (flaps-up or 10° landing with residual ice)**

Deferred: Residual airframe ice – flaps-up / 10° landing at increased speed, longer runway; extend gear early. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff

Power loss on the runway or just airborne

▶ Airspeed and situation?

Below 75 KIAS – on runway:

- Throttles (memory) ..... **CLOSED (both)**
- Brakes (memory) ..... **APPLY, stop straight ahead**
- Mixtures ..... **IDLE CUT-OFF**
- Magnetos / master ..... **OFF**
- Fuel selectors ..... **OFF**

Above 75 KIAS – runway remains:

- Throttles (memory) ..... **CLOSED (both)**
- Land / stop (memory) ..... **STRAIGHT AHEAD, maximum braking**

Below approx 500 ft AGL with runway remaining, landing is preferable to continuing on one engine.

Airborne – cannot stop on runway:

Continue with Engine Failure After Liftoff.

Engine Failure After Liftoff

Airborne, one engine failed

LAND AS SOON AS POSSIBLE

- Directional control (memory) ..... **MAINTAIN (rudder toward good engine, bank 2–5° into good engine)**
- Airspeed (memory) ..... **88 KIAS (Vyse), minimum 82 KIAS (Vxse)**

▶ Gear position?

Gear DOWN – runway / area ahead:

- Throttles (memory) ..... **CLOSED (both)**
- Land (memory) ..... **STRAIGHT AHEAD on remaining runway or suitable area**
- Flaps ..... **AS REQUIRED**
- Mixtures / mags / master ..... **OFF before touchdown**

Gear UP or no landing area:

- Mixtures (memory) ..... **FULL RICH**
- Propeller controls (memory) ..... **FULL FORWARD**
- Throttles (memory) ..... **FULL FORWARD**
- Flaps (memory) ..... **UP**
- Landing gear (memory) ..... **UP**
- Inoperative engine (memory) ..... **IDENTIFY (dead foot = dead engine)**
- Inoperative engine (memory) ..... **VERIFY (close its throttle, no change)**
- Inoperative engine (memory) ..... **FEATHER (prop control full aft)**
- Airspeed (memory) ..... **88 KIAS, climb**
- Trim ..... **SET (rudder trim into good engine)**
- Inoperative engine ..... **SECURE (see Engine Securing)**
- Operating engine ..... **MONITOR, cowl flap OPEN, electric fuel pump ON**
- Deferred: One engine inoperative – single-engine approach and landing; no go-around once gear and flaps are down. ..... **NOTED**

Climb performance on one engine is marginal; do not turn or climb with gear down unless necessary. Avoid banking into the dead engine.

Engine Failure in Flight

One engine loses power with altitude available

LAND AS SOON AS POSSIBLE

- Directional control (memory) — **MAINTAIN (rudder, bank into good engine)**
- Airspeed (memory) — **88 KIAS (Vyse) or above**
- Mixtures (memory) — **FULL RICH**
- Propeller controls (memory) — **FULL FORWARD**
- Throttles (memory) — **FULL FORWARD**
- Flaps / gear (memory) — **UP**
- Inoperative engine (memory) — **IDENTIFY (dead foot = dead engine)**
- Inoperative engine (memory) — **VERIFY (close its throttle, no change)**

Below Vmc (56 KIAS): reduce power on operating engine and lower the nose to regain airspeed and directional control.

▶ Altitude and time to attempt a restart?

Yes – troubleshoot:

- Fuel selector (inop) — **CHECK ON (or X-FEED)**
- Electric fuel pump (inop) — **ON**
- Mixture (inop) — **FULL RICH**
- Carb heat (inop) — **ON**
- Magnetos (inop) — **CHECK BOTH**
- Oil pressure / fuel pressure (inop) — **CHECK**

▶ Engine running?

Yes:

- Power — **REDUCE to minimum required**
- Engine instruments — **MONITOR**
- Deferred: Engine restored after power loss – land as soon as possible; be ready to feather. — **NOTED**

No:

- Inoperative engine — **FEATHER (prop control full aft)**
- Inoperative engine — **SECURE (see Engine Securing)**

No – feather now:

- Inoperative engine (memory) — **FEATHER (prop control full aft)**
- Inoperative engine — **SECURE (see Engine Securing)**

- Trim — **SET (rudder trim into good engine)**
- Operating engine — **POWER AS REQUIRED, cowl flap OPEN, monitor CHT / oil temp**
- Electrical load — **REDUCE (one alternator)**
- Fuel — **X-FEED from inop side if required (level flight only)**

Deferred: One engine inoperative – single-engine approach and landing; land at nearest suitable airport. — **NOTED**

Engine Securing (After Feather)

Engine feathered, aircraft under control

- Throttle (inop) — **CLOSED**
- Mixture (inop) — **IDLE CUT-OFF**
- Propeller control (inop) — **FEATHER (confirm)**
- Fuel selector (inop) — **OFF**
- Electric fuel pump (inop) — **OFF**
- Magnetos (inop) — **OFF**
- Alternator (inop) — **OFF**
- Cowl flap (inop) — **CLOSED**
- Electrical load — **REDUCE to essential**
- Operating engine — **Electric fuel pump ON, cowl flap OPEN as required, mixture as required**
- Fuel crossfeed — **AS REQUIRED (X-FEED on operating side to use inop-side fuel, level flight)**
- Airspeed — **88 KIAS minimum**
- Trim — **SET**
- ATC — **DECLARE EMERGENCY, nearest suitable airport**

Single-Engine Approach and Landing

One engine feathered / secured

Seat belts / harnesses	**FASTENED**
Fuel selector (operating)	**ON (crossfeed OFF for landing)**
Fuel selector (inop)	**OFF**
Electric fuel pump (operating)	**ON**
Mixture (operating)	**FULL RICH**
Propeller (operating)	**FULL FORWARD**
Cowl flap (operating)	**AS REQUIRED**
Approach	**LONG, SHALLOW final; power as required**
Airspeed	**90 KIAS on final (88 KIAS minimum until landing assured)**
Landing gear	**DOWN when landing is assured (below 140 KIAS), three green**
Flaps	**25° when landing is assured; 40° only on short final**
Airspeed	**REDUCE to 80 KIAS over the threshold**
Throttle	**REDUCE smoothly in the flare (expect yaw change)**

Do not attempt a single-engine go-around once gear and full flaps are down below approx 500 ft AGL.

Single-Engine Go-Around

Avoid if at all possible; only with altitude and airspeed in hand

Mixture (operating) (memory)	**FULL RICH**
Propeller (operating) (memory)	**FULL FORWARD**
Throttle (operating) (memory)	**FULL FORWARD**
Flaps (memory)	**UP (retract to 25°, then UP as speed allows)**
Landing gear (memory)	**UP (positive rate)**
Airspeed (memory)	**88 KIAS (Vyse)**
Directional control	**RUDDER, bank 2–5° into good engine**
Trim	**SET**

Expect little or no climb capability at high weight or density altitude.

Air Start / Unfeathering

Engine secured, restart desired (fault corrected)

Fuel selector (inop)	**ON**
Electric fuel pump (inop)	**ON**
Throttle (inop)	**OPEN 1/4 inch**
Propeller control (inop)	**FORWARD to mid-range (out of feather)**
Mixture (inop)	**FULL RICH**
Magnetos (inop)	**ON (both)**
Alternator (inop)	**ON**

▶ Propeller unfeathers and windmills (accumulator)?

Yes:

Engine	**ALLOW to fire, adjust throttle**

No:

Starter (inop)	**ENGAGE until prop windmills / engine fires**
Throttle	**ADJUST approx 1,500 RPM**
Oil pressure	**CHECK rising**
Engine	**WARM UP at reduced power (approx 2,000 RPM) until oil temp in green**
Cowl flap (inop)	**OPEN**
Power	**INCREASE gradually, resynchronize**
Electric fuel pump	**OFF (check pressure)**

Deferred: Engine restarted in flight – land as soon as practical; keep power low on the affected engine. **NOTED**

Engine Fire in Flight

LAND IMMEDIATELY

- Fuel selector (affected) (memory) ..... **OFF**
- Throttle (affected) (memory) ..... **CLOSED**
- Propeller (affected) (memory) ..... **FEATHER**
- Mixture (affected) (memory) ..... **IDLE CUT-OFF**
- Cowl flap (affected) (memory) ..... **OPEN (purge)**
- Heater / defroster (memory) ..... **OFF**
- Electric fuel pump (affected) ..... **OFF**
- Magnetos / alternator (affected) ..... **OFF**
- Airspeed ..... **88 KIAS or higher to help extinguish**
- Single-engine landing ..... **EXECUTE at nearest suitable airport (do not attempt restart)**

Engine Fire During Start

- Starter (memory) ..... **CONTINUE CRANKING (draw fire into engine)**
- Mixture (memory) ..... **IDLE CUT-OFF**
- Throttle (memory) ..... **FULL OPEN**
- Electric fuel pump ..... **OFF**
- Fuel selector ..... **OFF**

▶ Fire continues?

Yes:

- Magnetos / master ..... **OFF**
- Aircraft ..... **EVACUATE, fire extinguisher**

No:

- Engine ..... **SHUT DOWN, inspect before further flight**

Electrical / Cabin Fire

LAND AS SOON AS POSSIBLE

- Master switch (memory) ..... **OFF**
- Alternators (memory) ..... **OFF**
- Avionics master (memory) ..... **OFF**
- Cabin heat / defroster (memory) ..... **OFF**
- Vents ..... **OPEN**
- Fire extinguisher ..... **USE (if fire in cabin)**

After discharge, ventilate the cabin before landing.

▶ Fire out and electrical power essential?

Yes:

- Circuit breakers ..... **CHECK, do not reset tripped breakers**
- Master / alternators ..... **ON**
- Electrical items ..... **ON one at a time, watch for smoke**
- Deferred: Electrical fire – faulted circuit isolated; verify gear operation, use emergency extension if needed. ..... **NOTED**

No:

Master off: gear must be extended with the emergency gear knob (free-fall); no gear lights. Flaps are manual.

- Deferred: Master off – emergency (free-fall) gear extension, no gear lights, no radio. ..... **NOTED**

Emergency Descent

- Throttles (memory) ..... **CLOSED**
- Propeller controls (memory) ..... **FULL FORWARD**
- Mixtures (memory) ..... **FULL RICH**
- Landing gear (memory) ..... **DOWN (below 140 KIAS)**
- Flaps (memory) ..... **UP**
- Airspeed (memory) ..... **140 KIAS (gear down) / 135 KIAS turbulent**

Emergency Gear Extension

Gear fails to extend with selector, or no three green

Master switch	**ON**
Gear circuit breaker	**CHECK IN**
Panel lights	**OFF (daytime, to check gear lights)**
Gear light bulbs	**CHECK (swap bulbs if one is out)**
Airspeed	**BELOW 100 KIAS**
Gear selector	**DOWN**
Emergency gear extend knob	**PULL (releases hydraulic pressure, gear free-falls)**

▶ Three green?

Yes:

Emergency gear knob	**LEAVE PULLED, do not retract**
Deferred: Gear extended by free-fall – leave emergency knob out, do not retract gear, normal landing.	**NOTED**

No:

Aircraft	**YAW side to side (rudder) to help gear lock**
Airspeed	**REDUCE further (approach stall margin)**

▶ Three green now?

Yes:

Deferred: Gear extended by free-fall after yawing – leave emergency knob out, normal landing. **NOTED**

No:

Proceed to Gear-Up / Partial Gear Landing.

Gear-Up / Partial Gear Landing

Gear cannot be extended or locked

Runway	**LONGEST available, emergency services notified**
Fuel	**BURN DOWN to minimum practical**
Seat belts / harnesses	**TIGHT**
Loose items	**SECURE**

▶ Gear condition?

All up:

Gear selector	**UP**

Partial – one or two legs:

Gear	**RETRACT if possible; otherwise land on extended wheels, hold unsafe side off with aileron**
Approach	**NORMAL, flaps as desired, power on both engines to the flare**
Electric fuel pumps	**OFF**
Mixtures	**IDLE CUT-OFF (short final, landing assured)**
Magnetos	**OFF**
Fuel selectors	**OFF**
Master / alternators	**OFF**
Door	**UNLATCH**
Touchdown	**MINIMUM speed, nose slightly high, then full aft**
Aircraft	**EVACUATE when stopped**